

Vitamin D Deficiency & Rickets in Children

Clinical Summary for Physicians — based on UpToDate: Misra M, et al. "Vitamin D insufficiency and deficiency in children and adolescents" (Jun 2020)

1. DEFINITION & METABOLISM

Vitamin D is a prohormone essential for calcium/phosphorus homeostasis and bone mineralization. Severe deficiency causes **rickets** (growing children — failure of growth-plate/bone mineralization) and **osteomalacia** (any age — impaired mineralization of already-formed bone). Mild/chronic deficiency is linked to reduced bone mineral density even without overt rickets.

Key Forms of Vitamin D

- **Vitamin D3 (Cholecalciferol)** — made in skin by UVB; found in animal products/supplements.
- **Vitamin D2 (Ergocalciferol)** — from plant sources; found in most supplements.
- **Calcidiol (25-OH-D)** — storage form made in the **liver**; the best lab marker of vitamin D status.
- **Calcitriol (1,25-(OH)₂-D)** — active hormone made in the **kidney**; drives intestinal calcium absorption, bone resorption, and renal calcium/phosphate handling.

Pathway: Skin (UVB) or Diet → Cholecalciferol/Ergocalciferol → Liver (25-OH-D, storage) → Kidney (1,25-(OH)₂-D, active) → ↑ gut Ca absorption, ↑ bone resorption, ↓ renal Ca/PO₄ loss.

2. DIAGNOSTIC CLASSIFICATION (based on serum 25-OH-D)

Category	25-OH-D (ng/mL)	25-OH-D (nmol/L)
Sufficiency	20 – 100	50 – 250
Insufficiency	12 – 20	30 – 50
Deficiency	< 12	< 30

Recommended daily intake: Infants (0–12 mo) = 400 IU/day; Children/adolescents (1–18 yr) = 600 IU/day (higher in high-risk groups, up to 2000–6000 IU/day).

Vitamin D deficiency is common in infants who have dark skin pigmentation and those who are exclusively breastfed beyond three to six months of age, particularly if there are additional risk factors such as maternal vitamin D deficiency during pregnancy or prematurity

The third trimester is a critical time for vitamin D transfer

The vitamin D content of breast milk is low (15 to 50 international units/L [0.4 to 1.2 micrograms/L]) even in a vitamin D-sufficient mother.

body's compensation mechanism

3. STAGES OF VITAMIN D DEFICIENCY (Biochemical Progression)

Stage	Ca	PO4	ALP	PTH	25-OH-D	1,25-(OH) ₂ -D	X-ray
Early	N/↓	N/↓	↑	↑	↓	N	Reduced mineralization
Moderate	N/↓	↓	↑↑	↑↑	↓↓	↑	Rachitic changes +
Severe	↓↓	↓↓	↑↑↑	↑↑↑	↓↓↓	↑/N/↓	Rachitic changes ++

Pathophysiology in one line: ↓25-OH-D → ↓gut Ca/PO₄ absorption → ↑PTH (mobilizes bone Ca, wastes PO₄ renally) → low Ca×PO₄ product → defective mineralization → rickets/osteomalacia.

4. CLINICAL MANIFESTATIONS

Most patients are **asymptomatic**. When symptomatic:

Skeletal (Rickets — growing children)

Vitamin D deficiency causes rickets in growing children and osteomalacia in all age groups.

- Delayed fontanelle closure, craniotables, frontal bossing
- Widened wrists/ankles, rachitic rosary (costochondral prominence)
- Bow legs (genu varum) or knock knees (genu valgum)
- Motor delay, poor growth, bone pain/irritability
- X-ray: metaphyseal widening/splaying/cupping/fraying, loss of provisional calcification zone

parathyroid hormone (PTH) and alkaline phosphatase levels increase

Osteomalacia (older adolescents/adults — growth complete)

- Diffuse or localized bone and muscle pain/weakness (no deformity, since growth plates are fused)

Severe / Hypocalcemic Manifestations

- **Muscle weakness**, difficulty standing/walking (from hypophosphatemia)
- **Tetany**, **seizures**, **apnea**, **stridor**, **wheezing**, **hypotonia**, hyperreflexia (from severe hypocalcemia — most common in infancy and adolescence, periods of rapid growth)

Vitamin D ↓ → intestinal Ca absorption ↓ + phosphate absorption ↓

Both needed to make strong, mineralized bone

PTH tries to maintain blood calcium
Calcium is pulled from bone

But PTH also causes phosphate loss in urine
↓
Blood phosphate ↓↓↓

Poor mineralization of growing bone
↓
Rickets

ALP is an enzyme associated with bone formation

5. EPIDEMIOLOGY

- US pediatric prevalence of deficiency/insufficiency (25-OH-D **<20 ng/mL**): ~15%; severe deficiency (**<10 ng/mL**) in 1–2%.
- Europe: ~45% of children/adolescents have insufficiency or deficiency.
- Highest rates of rickets: **Black children** > Asian Indian children > White children — tracking with skin pigmentation.
- Black non-Hispanic adolescents have ~20× the risk of low 25-OH-D vs White non-Hispanic adolescents (NHANES III); risk higher in females.
- Deficiency is far more common at end of winter than end of summer; more common at latitudes beyond 40°.

6. ETIOLOGY & RISK FACTORS

Perinatal / Infant Risk Factors

- **Maternal vitamin D deficiency** during pregnancy (crosses placenta; critical in 3rd trimester for fetal skeletal calcification)
- **Prematurity** — less time for transplacental transfer
- **Exclusive breastfeeding beyond 3–6 months** without supplementation — breast milk vitamin D content is low (major cause: >95% of US rickets cases were in breastfed infants)
- **Dark maternal/infant skin pigmentation**

Common Postnatal Risk Factors (Older Children)

- **Dark skin pigmentation** (melanin blocks UVB synthesis; darker skin needs **3–10×** more sun exposure)
- **Low dietary intake** — limited dairy, vegan/unusual diets, poor fortified-food intake
- **High latitude residence**, winter season, limited sun exposure, consistent sunscreen use, staying mostly indoors
- **Obesity** — vitamin D sequestered in fat tissue → higher requirement

Medications & Medical Conditions

- **Anticonvulsants**, antiretrovirals (increase 25-OH-D catabolism)
- **Chronic glucocorticoids** (impair intestinal Ca absorption)
- **Ketoconazole/antifungals** (block 1-hydroxylation)
- **Malabsorptive conditions**: celiac disease, IBD, cystic fibrosis (pancreatic insufficiency), cholestatic liver disease, post gut-resection/bariatric surgery
- Chronic kidney disease, severe hepatic dysfunction

Genetic Causes (rare)

- 25-hydroxylase deficiency (CYP2R1 mutation)
- 1-alpha-hydroxylase deficiency (CYP27B1 mutation) — low 1,25-(OH)₂-D, normal 25-OH-D
- Hereditary vitamin D resistance (VDR gene mutation) — alopecia + normal/high 25-OH-D and 1,25-(OH)₂-D

Note: In nephrotic syndrome, low total 25-OH-D may reflect low vitamin D-binding protein (free/bioavailable 25-OH-D may be normal) — not true deficiency.

7. WHOM & HOW TO TEST (Screening)

Screen only at-risk groups (not the general population):

- Exclusively breastfed/premature infants not reliably taking vitamin D supplements
- Dark-skinned children at higher latitudes (screen in winter/spring when levels are lowest)
- Children with low dietary vitamin D intake and no supplementation
- Infants/children with unexplained poor growth, motor delay, or irritability
- Children on anticonvulsants, antiretrovirals, or chronic glucocorticoids
- Chronic malabsorptive disease, low bone density conditions, CKD, or severe liver disease
- Pregnant or lactating adolescents
- Markedly elevated alkaline phosphatase for age (>500 IU/L in neonates; >1000 IU/L in children <9 yr)

(Screening in obesity alone is controversial/optional.)

8. INVESTIGATIONS

- Serum 25-OH-D — the test of choice for vitamin D status (best reflects total body stores; half-life 2–3 weeks)
- 1,25-(OH) $_2$ -D is not useful for screening (short half-life, PTH-driven fluctuations, may be normal/high even in deficiency)
- If 25-OH-D <20 ng/mL: add serum calcium, phosphorus, alkaline phosphatase, and PTH
- Radiographs (wrist/knee) if age <3 years or high clinical suspicion of rickets
- Classify rickets as calcipenic (hypocalcemic) vs hypophosphatemic based on labs

9. DIAGNOSTIC THRESHOLDS (Recap)

Sufficiency ≥ 20 ng/mL (50 nmol/L) · Insufficiency 12–20 ng/mL (30–50 nmol/L) · Deficiency <12 ng/mL (<30 nmol/L).
Radiologic rickets/low bone density appear below ~ 16 –18 ng/mL; ALP begins rising below ~ 20 ng/mL.

10. MANAGEMENT (Treatment)

Treat if 25-OH-D <20 ng/mL (50 nmol/L) or clinical/radiologic rickets. Either D2 or D3 may be used.

Group	Treatment dose	Duration → Maintenance
Infants <12 mo	2000 IU/day	6–12 wk → then ≥400 IU/day
Children ≥12 mo (deficiency, no rickets)	2000 IU/day (or 50,000 IU once weekly ×6 wk)	6–12 wk → then 600–1000 IU/day
Established rickets, 12 mo–12 yr	3000–6000 IU/day	12 wk → then maintenance
Established rickets, ≥12 yr	6000 IU/day	12 wk → then maintenance
Obesity / malabsorption / interacting medications	2–3× standard dose (up to 6000 IU/day)	Higher maintenance dose

Calcium Co-supplementation

- Give if elevated PTH or rickets present — prevents "hungry bone syndrome" hypocalcemia during repletion
- Elemental calcium 30–75 mg/kg/day (divided 2–3×/day) for 2–4 weeks, or per Global Consensus: 500 mg/day elemental calcium
- Severe symptomatic hypocalcemia (seizures/tetany): IV calcium gluconate 10–20 mg/kg elemental Ca (max 540 mg/dose), slow IV

Monitoring / Follow-up

- Check calcium after 1–2 months in all infants and anyone on >2000 IU/day (risk of hypercalcemia, esp. CYP24A1 mutations)
- Uncomplicated low 25-OH-D: recheck at 2–3 months
- Elevated ALP/PTH without rickets: recheck at 6–8 weeks, then periodically, then annually
- Rickets: close follow-up for radiographic healing and normalization of labs

11. PREVENTION

- **All exclusively breastfed infants:** 400 IU/day vitamin D starting within days of birth, continued until weaned onto ≥1 L/day of vitamin D-fortified formula or milk
- Formula-fed infants consuming <1 L/day of fortified formula also need supplementation
- Alternative: high-dose maternal supplementation (4000–6400 IU/day) can adequately enrich breast milk
- Pregnant women: ensure adequate intake (some data suggest >600 IU/day needed, especially in dark-skinned women)
- Older children: encourage vitamin D-fortified dairy/foods; routine sun exposure is NOT recommended as a prevention strategy (skin cancer risk)

12. KEY TAKE-HOME POINTS

- Rickets = pediatric mineralization failure at growth plates; osteomalacia = mineralization defect after growth plates fuse.
- 25-OH-D is the correct screening test — not 1,25-(OH)₂-D.
- Screen only high-risk groups; treat if 25-OH-D <20 ng/mL or rickets present.
- Give calcium alongside vitamin D when PTH is elevated or rickets is present, to avoid hungry-bone hypocalcemia.
- Prevention (400 IU/day for breastfed infants; 600 IU/day for children) is simple, safe, and effective.

Source: Misra M. "Vitamin D insufficiency and deficiency in children and adolescents." UpToDate, literature review current through May 2020, topic last updated Jun 11, 2020. This is a condensed educational summary — refer to the original topic review and local guidelines for full clinical detail and dosing verification.